

ANUPAMA SINGH

Software Engineer — Backend & Platform

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SUMMARY

Full-stack software engineer with 4+ years of professional experience building scalable web applications, backend services, and developer tooling across React, Angular, Node.js, and Spring Boot. Hands-on with Kubernetes, Docker, REST APIs, and CI/CD on cloud platforms (Azure, AWS). Currently pursuing M.S. in Software Engineering at San Jose State University with coursework in Distributed Systems, Machine Learning, and Cloud Computing. Interested in building reliable platform services, operational tooling, and AI-assisted workflows that scale to large user bases.

EDUCATION

San Jose State University — M.S. in Software Engineering *Aug 2025 – May 2027 (Expected)*
GPA: 3.82 / 4.0 *San Jose, CA*

Relevant Coursework: Distributed Systems, Cloud Computing, Machine Learning, Artificial Intelligence, Data Engineering, Database Systems, Data Structures & Algorithms, Operating Systems.

Oriental College of Technology (RGPV) — B.E., Electronics & Communication *2015 – 2019*
CGPA: 8.02 / 10 *Bhopal, India*

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Java, C++, HTML, CSS, SQL

Frontend: React.js, Angular 14, Shadcn/UI, jQuery, Thymeleaf, Progressive Web Apps

Backend & APIs: Node.js, Spring Boot, Next.js, REST APIs, Microservices, Swagger, Postman

Cloud & DevOps: AWS (CloudFormation, CodePipeline, CodeCommit), Azure, Docker, Kubernetes, CI/CD, Git, Gradle

Data & Observability: MySQL, PostgreSQL, MongoDB, Prometheus, Grafana, Kafka

ML / AI: Amazon Lex, Dialogflow, NLP, XGBoost, SHAP (explainable ML), RAG systems

PROFESSIONAL EXPERIENCE

Tata Consultancy Services — Tech Lead, Full-Stack Engineer *Jan 2024 – Aug 2025*
Pune, India

- Led delivery of 10+ platform features on a Node.js (SSBT) runtime, improving application performance and stability for a large internal user base.
- Designed and shipped a full-stack Spring Boot service with scheduled batch jobs to transform encrypted data, driving a 68% increase in account-creation throughput.
- Cut page reload time by 45% by re-architecting UI workflows and introducing client- and service-level caching strategies.
- Improved deployment reliability by 25% by debugging and tuning Kubernetes workloads on Azure, including resource limits, probes, and rolling-update strategies.
- Built automation around file ingestion and search using advanced regex pipelines, eliminating ~7 hours per week of manual operational work.

Tata Consultancy Services — Software Engineer, Full-Stack *Nov 2022 – Dec 2023*
Pune, India

- Developed an Angular 14 single-page application integrating 5 backend microservices, exposing unified APIs and centralized operational visibility.
- Designed and tested REST APIs using Postman and Swagger, ensuring reliable service-to-service communication and contract stability.

- Built a reusable dynamic form-builder (Google Forms-style) on React + Shadcn/UI, reducing feature delivery time by 30% for downstream teams.
- Modernized legacy frontends to current ECMAScript standards, delivering a defect-free release and reducing long-term maintenance cost.

Accenture — Application Development Analyst

Jan 2020 – Oct 2022

Pune, India

- Built Angular + Spring Boot CI/CD pipelines that cut deployment time by 90% and reduced release-related incidents.
- Developed AI chatbots on Amazon Lex and Dialogflow to automate internal user interactions and triage workflows — directly relevant to AI-assisted operational tooling.
- Designed an NLP-based PoC using neural networks that reduced inbound support ticket volume by 27% through intent classification and auto-resolution.
- Shipped an Electron-based desktop application that reduced data synchronization time by 98% for field users.

SELECTED PROJECTS

Clinical Trials Management Platform — Distributed Systems (SJSU)

- Designed an enterprise-grade platform supporting end-to-end clinical-trial lifecycle with modular services, RESTful APIs, and role-based access control over sensitive data.
- Integrated Prometheus + Grafana for metrics, dashboards, and observability across services — patterns directly applicable to telemetry/safety platforms.

AWS Cloud Infrastructure Challenge — Cloud Computing (SJSU)

- Diagnosed and remediated CloudFormation templates flagged by cfn_nag (missing SecurityGroupEgress, missing Description) and cfn-lint (deprecated Python runtime).
- Delivered fixes through a CodeCommit + CodePipeline workflow, reinforcing infrastructure-as-code, CI/CD, and policy-driven security review.

Distributed Chord — Task Execution Engine (SJSU)

- Implemented the Chord distributed hash table protocol to route and execute tasks across a peer-to-peer ring, including finger-table maintenance, successor/predecessor stabilization, and graceful node join/leave.
- Built fault-tolerant task dispatch over the ring with consistent hashing for key partitioning, demonstrating $O(\log N)$ lookup latency under churn — patterns directly applicable to scalable platform services.

Explainable Risk Classification — Machine Learning (SJSU)

- Team project on California HMDA mortgage data using XGBoost with SHAP-based explainability, producing per-decision feature attributions suitable for policy/audit review.
- Authored the IEEE-format conference paper and delivered model evaluation across precision, recall, and confusion-matrix metrics.

ACHIEVEMENTS

- On-the-Spot Team Award, TCS (Jan 2024); Xcelerate Warrior Appreciation, TCS (Dec 2023).
- Client Deliverables Excellence Award, Accenture (May 2022).